

Mt. Stony Brook Observing Proposal

Date: March 13, 2026

Panel: *For office use.*

Category: Star Clusters

Determining the Age of the Open Cluster M67 via BV Photometry and Isochrone Fitting

PI: 114956855

Abstract of Scientific Justification (*will be made publicly available for accepted proposals*):

Open star clusters are groups of stars that all formed from the same gas cloud at the same time. Because every star in the cluster shares the same age and distance from Earth, we can use their collective brightness and colors to figure out how old the cluster is. We propose to image the open cluster M67 (NGC 2682) in two filters (B and V) using the Mt. Stony Brook 14-inch telescope. From these images we will measure the brightness and color of hundreds of cluster stars and build a color-magnitude diagram (CMD), a plot of stellar brightness versus color. We will then compare the shape of the CMD to computer-generated stellar evolution models called isochrones, which predict what a cluster of a given age should look like. M67 is an ideal target because its stars are at convenient brightnesses for our telescope ($V \approx 10\text{--}16$), it is well-placed in the sky, and its ~ 4 Gyr age produces a clear, distinctive CMD shape that makes age determination straightforward. Our quantitative goal is to determine the cluster age to within ± 0.5 Gyr and compare our result to the accepted literature value.

Summary of observing runs requested for this project

Run	Telescope	Instrument	Time Request	Moon	Optimal months	Accept. months
1	14-inch	imaging	2x 2 hours	dark	Mar - Apr	Mar - May
2						
3						
4						
5						
6						

Scheduling constraints and non-usable dates (*up to six lines*).

No constraints beyond lunar phase and target visibility.

Scientific Justification Be sure to include overall significance to astronomy. For standard proposals limit text to one page with figures, captions and references on no more than two additional pages.

Background and Motivation. Stars spend most of their lives fusing hydrogen in their cores on the “main sequence.” More massive stars burn through their fuel faster and leave the main sequence sooner, so the shape of a cluster’s color–magnitude diagram (CMD), a plot of stellar brightness versus color, changes predictably with age. By comparing an observed CMD to theoretical curves called isochrones, which show what a cluster of a given age should look like, we can directly measure how old the cluster is [Dotter, 2016]. The key feature we look for is the main-sequence turnoff (MSTO): the point where the most massive remaining stars are just beginning to leave the main sequence. The brighter and bluer this turnoff, the younger the cluster.

Target: M67. M67 (NGC 2682; $\alpha = 08^{\text{h}}51^{\text{m}}18^{\text{s}}$, $\delta = +1148'$, J2000) is one of the most well-studied open clusters in the northern sky. At a distance of about 908 pc (~ 3000 light-years) and an age of roughly 4 Gyr [Richer et al., 1998, Sarajedini et al., 2009], it is old enough to show a clear, dramatic turnoff and a population of evolved red giant stars.

Science Goal. We will build a CMD of M67 from our B and V images and find the best-matching isochrone from the PARSEC stellar evolution library [Bressan et al., 2012]. Our quantitative goal is to determine the cluster age to within ± 0.5 Gyr and compare our result to the accepted value of 3.5–4.0 Gyr.

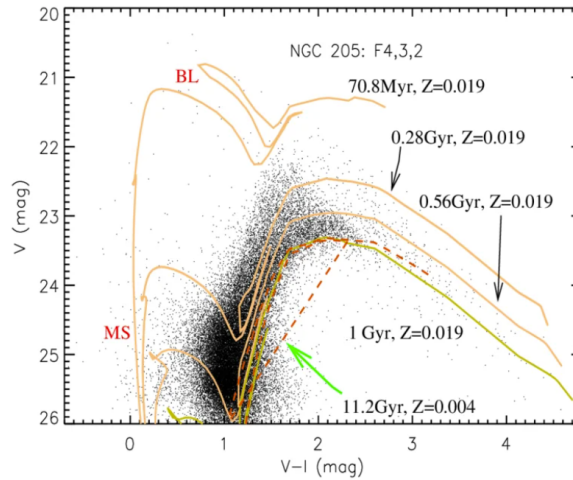


Figure 1: An example isochrone fitting of NGC 205 stars. The filters used here are V and I , this image is from [Butler and Delgado, 2005]. Multiple isochrones representing different ages are overplotted.

References

- A. Bressan, P. Marigo, L. Girardi, B. Salasnich, C. Dal Cero, S. Rubele, and A. Nanni. PARSEC: stellar tracks and isochrones with the PAdova and TRieste Stellar Evolution code. *Monthly Notices of the Royal Astronomical Society*, 427:127–145, 2012. doi: 10.1111/j.1365-2966.2012.21948.x.
- D. Butler and D. Delgado. On the stellar populations in ngc 185 and ngc 205 and the nuclear

- star cluster in ngc 205 fromhubble space telescopeobservations. *The Astronomical Journal*, 129: 2217–2231, 05 2005. doi: 10.1086/429524.
- A. Dotter. MESA isochrones and stellar tracks (MIST) 0: Methods for the construction of stellar isochrones. *The Astrophysical Journal Supplement Series*, 222:8, 2016. doi: 10.3847/0067-0049/222/1/8.
- H. B. Richer, G. G. Fahlman, J. Rosvick, and R. Ibata. The white dwarf cooling age of M67. *The Astrophysical Journal Letters*, 504:L91–L95, 1998. doi: 10.1086/311586.
- A. Sarajedini, A. Dotter, and A. Kirkpatrick. Deep ACS imaging of the galactic open clusters NGC 6791 and NGC 6819: Constraining models of stellar evolution. *The Astrophysical Journal*, 698: 1872–1878, 2009. doi: 10.1088/0004-637X/698/2/1872.

Experimental Design Describe your overall observational program. How will these observations contribute toward the accomplishment of the goals outlined in the science justification? If you've requested long-term status, justify why this is necessary for successful completion of the science. (limit text to one page)

Target and Observability. M67 is located at $\alpha = 08^{\text{h}}51^{\text{m}}$, $\delta = +1148'$ (J2000). From Stony Brook (latitude $+40.9\text{N}$), M67 rises to a maximum altitude of $\sim 61^\circ$ above the horizon. During March–April, M67 is highest in the sky around 9–10 PM EDT, giving us a comfortable 4–5 hour observing window each night. Our telescope's field of view is about $22.8' \times 15'$, which covers the dense central region of M67. Although this does not capture the full 30 arcmin extent of the cluster, the central core contains hundreds of member stars, more than enough to populate all the key CMD features we need for isochrone fitting.

Filter Choice. We will observe in the B (blue) and V (visual/green) filters. Using two filters lets us measure both the brightness and the color of each star. Color is the key for a CMD: it tells us how hot a star is, which in turn tells us where it sits on the main sequence. PARSEC isochrone models are tabulated directly in B and V .

Exposure Time Estimate. We need enough light from each star to measure its brightness accurately (signal-to-noise ratio $\text{SNR} \gtrsim 30$). Because M67 members range widely in brightness ($V \approx 10\text{--}16$), we use two exposure lengths per filter:

- *Short exposures* (30 s in B and V): For the brighter members ($V \approx 10\text{--}13$). At $V = 13$, a 30 s exposure should be well above the noise and safely below the saturation limit. If any stars are found to be saturated, exposure time will be reduced to 10 s.
- *Long exposures* (300 s in B and V): For the fainter members ($V \approx 13\text{--}16$). Stars at $V = 16$ are about 16 times fainter than those at $V = 13$;

We will take 3 exposures at each setting so that we can combine them to remove cosmic rays. Total science time per night: $2 \text{ filters} \times 2 \text{ exptimes} \times 3 \text{ exposures} \times (30 + 300) \text{ s} \approx 66$ minutes. Adding time for focusing and readout, each observing session takes about 100 minutes. We request 2 nights to ensure at least one clear, photometric night.

Calibration. Each night we will take: (1) 3 bias frames to measure the electronic zero-point of the CCD; (2) 10 flat fields in each filter to correct for uneven sensitivity across the detector; and (3) images of 2–3 star fields at different altitudes, which let us convert our raw instrumental brightnesses into calibrated magnitudes on the standard B/V system.

Measurement Plan. We will measure stellar brightnesses from our images using aperture photometry in Python or Source Extractor. After calibration and membership selection, we will plot the CMD and overlay PARSEC isochrones of different ages (2–6 Gyr in steps of 0.25 Gyr) at the known distance and reddening of M67. The best-fitting isochrone, the one whose turnoff point best matches the observed CMD, gives our age measurement.

Analysis Plan Describe the steps that you will take to analyze the data. State which tasks will be due for the weekly check-ins (weeks 1, 2, 3). The week 4 task will be to complete the lab report. In case you have less than four weeks between taking the observations and the last day to submit lab reports, how will the analysis plan change?

- +1 week: Reduce raw images: subtract bias frames, apply flat-field correction, and stack the 3 exposures per setting to remove cosmic rays. Run aperture photometry in SExtractor and Python on all detected stars in both B and V ; use Landolt standard star observations to convert raw counts into calibrated magnitudes.
- +2 weeks: Plot the color-magnitude diagram (V vs. $B - V$) for cluster members.
- +3 weeks: Overlay PARSEC isochrones of different ages (2–6 Gyr) on the CMD. Identify the best-fitting isochrone by eye and by minimizing scatter at the main-sequence turnoff. Estimate the age uncertainty from the range of isochrones that plausibly fit the data. Compare our result to the literature value.
- +4 weeks: Submit the lab report.

Observing Run Details for Run 1: 14-inch/ ZWO ASI2600MM DUO

Technical Description Describe the observations to be made during this observing run. Justify specific instrument choices such as the required filter, or the grating choice, as well as the requested lunar phase. List targets, coordinates, and magnitudes (or surface brightness, if appropriate) in the Target Tables section below.

Run 1 uses the Mt. Stony Brook 14-inch telescope with the ZWO ASI2600MM DUO imager in B and V filters. Two observing sessions of ~ 2 hours each are requested. Dark conditions are preferred to keep sky background low in the B band and reach the fainter cluster members ($V \approx 16$) with good signal-to-noise. Our field of view ($22.8' \times 15'$) covers the dense central core of M67, which contains sufficient member stars to populate all the CMD features needed for isochrone fitting.

Instrument Configuration

Filters: B,V
Grating/grism: -
Order: -
Cross disperser: -

Slit: -
Multislit: -
 λ_{start} : -
 λ_{end} : -

Fiber cable: -
Corrector: -
Collimator: -
Atmos. disp. corr.: -

R.A. range of principal targets (hours): 8 to 9

Dec. range of principal targets (degrees): 11 to 12

Special Instrument Requirements Describe briefly any special or non-standard usage of instrumentation.

Target Table for Run 1: 14-inch/STL-1001E

Obj ID	Object	α	δ	Epoch	Mag.	Filter	Exp. time	# of exp.	Lunar days	Sky	Seeing	Comment
001	M67	08:51:18	+11:48:00	2000.0	10-16	V	30	3	5	phot	3.0	short, bright members
002	M67	08:51:18	+11:48:00	2000.0	10-16	V	300	3	5	phot	3.0	long, faint members
003	M67	08:51:18	+11:48:00	2000.0	10-16	B	30	3	5	phot	3.0	short, bright members
004	M67	08:51:18	+11:48:00	2000.0	10-16	B	300	3	5	phot	3.0	long, faint members